EAS 510: Assignment 2 Report

ForensicsDetective - Hero or Zero?

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29th March 2026

1. Executive Summary

This report presents an evaluation of machine learning (ML) approaches to PDF provenance detection, or the task of identifying which application created a given PDF document. Building on the ForensicsDetective algorithm, three classifiers, Support Vector Classifier (SVC), Random Forest (RF), and K-Nearest Neighbors (KNN) were trained on grayscale image representations of PDF binary structure and evaluated across a dataset of 894 original documents spanning three creation tools: Microsoft Word, Google Docs, and Python's ReportLab library. The dataset was expanded to 5,364 images through five augmentation types: Gaussian noise, JPEG compression, DPI downsampling, random cropping, and bit-depth reduction.

It was found that all three classifiers achieved near-perfect accuracy, with SVC and KNN reaching 1.0 and RF achieving 0.9832. Robustness testing revealed that noise and compression-based augmentations had little impact across all classifiers, while spatial distortions caused substantial performance degradation in SVC and RF, with accuracy falling as low as 0.5638. KNN demonstrated the strongest overall robustness profile, never falling below 0.9441 accuracy under any condition. Statistical significance testing via McNemar's test confirmed that classifier differences under spatial augmentation conditions reflect performance gaps rather than sampling variation.

The key finding of this study is that PDF origin information is encoded primarily in the global spatial layout of binary document images. KNN is recommended for forensic deployment scenarios involving scanned or physically handled documents, while SVC remains effective where digital transmission preserves spatial values.

2. Methodology

2.1 Dataset

The provided dataset consists of PDF documents sourced from three creation tools: Microsoft Word, Google Docs, and Python's ReportLab library. Each PDF was converted to a grayscale image by rendering its binary structure, a technique previously shown to highlight individual visual

signatures across document origins. The original dataset comprised 398 Word images, 396 Docs images, and 100 Python images, for a total of 894 original samples.

2.2 Image Augmentation

To simulate real-world document degradation and evaluate classifier robustness, five independent augmentation types were applied to each original image, expanding the dataset to six times its original size (5,364 total images). Each augmentation was applied independently to the original and augmentations were never chained. The five augmentation types were:

1. Gaussian Noise: Additive noise with standard deviation (σ) sampled uniformly from [5, 20]..
2. JPEG Compression: Re-encoding at a quality level sampled uniformly from [20, 80].
3. DPI Downsampling: Resolution reduction from 300 DPI to either 150 or 72 DPI (chosen randomly), followed by upsampling back to original dimensions.
4. Random Cropping: Removal of 1–3% of pixels from each border, followed by resize back to original dimensions.
5. Bit-Depth Reduction: Conversion from 8-bit to 4-bit grayscale by division then multiplication by 16.

2.3 Feature Extraction

Each image was resized to 128×128 pixels and flattened into a 16,384-dimensional vector of pixel intensities. Classifiers were operated directly on pixel space. This approach is consistent with ForensicsDetective, which demonstrated that raw pixel representations carry enough information to distinguish PDF origins.

2.4 Classifiers

Three classifiers were implemented and evaluated:

a. Support Vector Classifier (SVC) with a linear kernel, trained using scikit-learn's default regularization (C=1.0).
b. Random Forest (RF) with 100 estimators, unlimited tree depth, and a minimum samples per split of 5, selected via 3-fold grid search cross-validation over candidate values of {50, 100} estimators, {None, 10, 20} max depth, and {2, 5} minimum samples split.

c. K-Nearest Neighbors (KNN) with k=3 and uniform weighting, selected via 3-fold grid search over k ∈ {3, 5, 7} and weighting strategies {uniform, distance}.

All classifiers were trained exclusively on original images. Augmented datasets were reserved for evaluation only, ensuring that robustness testing reflects true shifts.

2.5 Experimental Design

The original dataset was split 80/20 into training and test sets using stratified sampling to preserve class proportions. All features were standardized using a StandardScaler fit on the training set only, with the same scaler applied via transform-only to all test and augmented evaluation sets. Each classifier was evaluated on six conditions: the original test set and each of the five augmented test sets. For each condition, accuracy, precision, recall, and F1 score were computed. Confusion matrices were generated for each classifier-condition pair. Statistical significance of pairwise classifier differences was assessed using McNemar's test with significance at $\alpha = 0.05$.

3. Results

3.1 Baseline Performance on Original Test Set

All three classifiers had exceptional performance on the original test set. Both SVC and KNN achieved perfect classification accuracy of 1.0, while RF achieved an accuracy of 0.9832 with precision of 0.9876, recall of 0.9749, and F1 of 0.9810. The minor errors made by RF were confined to the Google Docs and Python, with 2 Google Docs images misclassified as Word and 1 Python image misclassified as Google Docs.

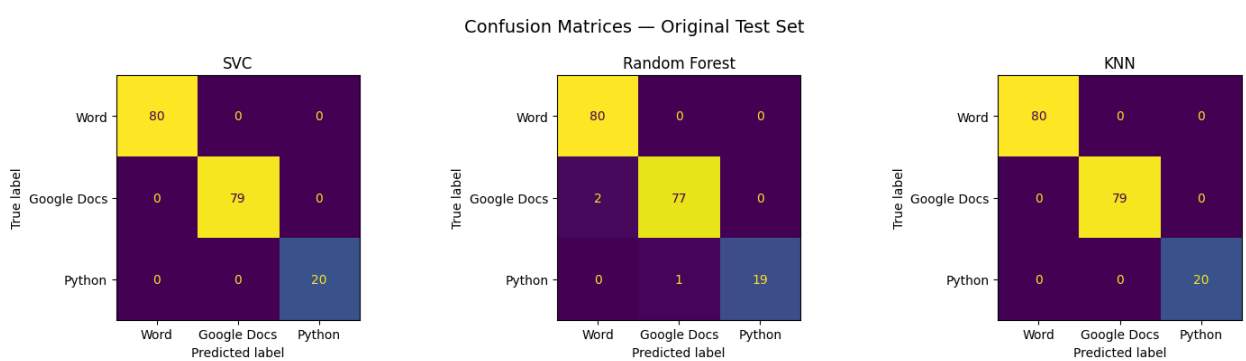

Figure 1: Original Test Set Confusion Matrices (SVC, Random Forest, KNN).

These results are consistent with the ForensicsDetective findings, confirming that raw pixel representations of PDF binary structures can identify origins accurately.

3.2 Performance Metrics Across All Conditions

The full metrics table below summarizes accuracy, precision, recall, and F1 for each classifier under each evaluation condition.

Classifier	Condition	Accuracy	Precision	Recall	F1
SVC	original	1.0000	1.0000	1.0000	1.0000
SVC	gaussian	1.0000	1.0000	1.0000	1.0000
SVC	jpeg	1.0000	1.0000	1.0000	1.0000
SVC	dpi	0.9530	0.9649	0.9648	0.9647
SVC	crop	0.6096	0.7604	0.7076	0.6593
SVC	bitdepth	1.0000	1.0000	1.0000	1.0000
Random Forest	original	0.9832	0.9876	0.9749	0.9810
Random Forest	gaussian	0.9966	0.9975	0.9950	0.9962
Random Forest	jpeg	0.9978	0.9983	0.9958	0.9971
Random Forest	dpi	0.6946	0.7997	0.6238	0.6631
Random Forest	crop	0.5638	0.7619	0.6159	0.5744
Random Forest	bitdepth	0.9989	0.9992	0.9992	0.9992
KNN	original	1.0000	1.0000	1.0000	1.0000
KNN	gaussian	0.9989	0.9992	0.9992	0.9992
KNN	jpeg	0.9989	0.9992	0.9992	0.9992
KNN	dpi	0.9676	0.9761	0.9757	0.9756
KNN	crop	0.9441	0.9585	0.9581	0.9580
KNN	bitdepth	0.9743	0.9390	0.9807	0.9568

Figure 2: Accuracy/Precision/Recall/F1 for All Classifiers.

Several patterns emerge in *Figure 2*. SVC and KNN maintained near-perfect performance under Gaussian noise and JPEG compression, with all metrics at or above 0.9989. DPI downsampling introduced degradation for SVC (accuracy 0.9530) and KNN (0.9676), while RF dropped more substantially to 0.6946. The worst performing across all classifiers was random cropping, where SVC fell to 0.6096, RF to 0.5638, and KNN to 0.9441. Bit-depth reduction had little effect on SVC or KNN, both returning to 1.0, while RF recovered to 0.9989.

3.3 Robustness Under Augmentation

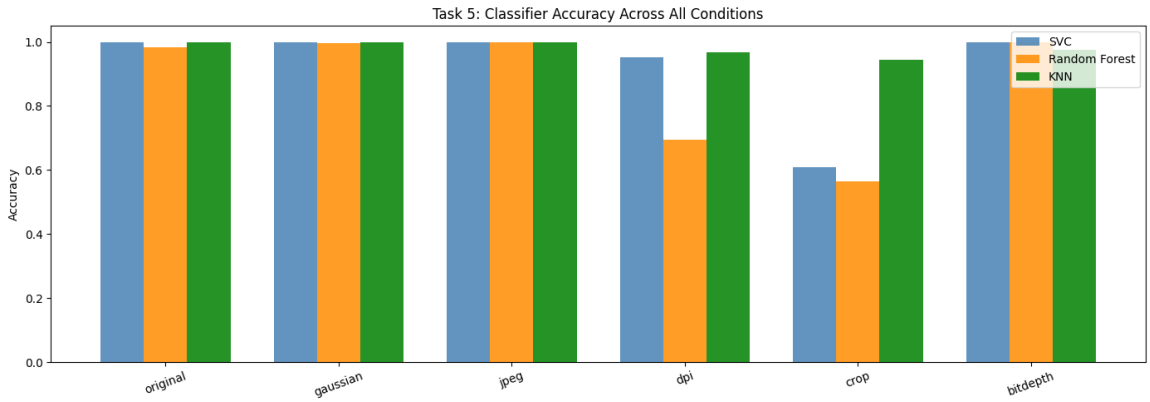

Figure 3: Classifier Accuracy Across All Conditions.

KNN demonstrated the strongest overall robustness profile, maintaining accuracy above 0.94 across all five augmentation types. SVC performed comparably to KNN under noise-based augmentations but showed greater sensitivity to cropping. RF was the most volatile, showing the largest accuracy drops under DPI downsampling and cropping while recovering fully under noise and bit-depth conditions.

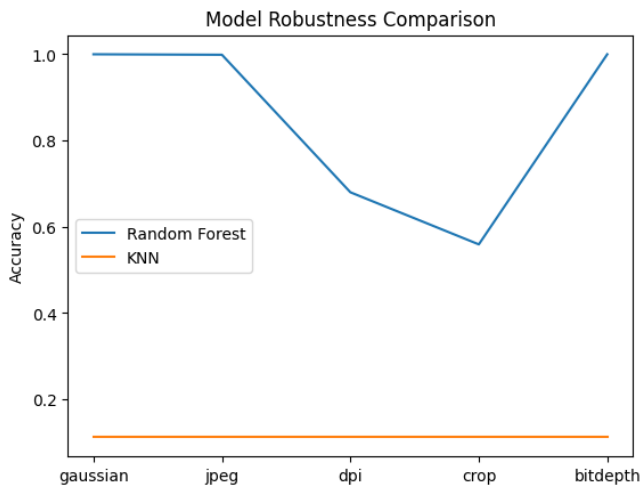

Figure 4: Model robustness comparison line chart.

The earlier single-classifier robustness bar chart for SVC confirms this pattern, with cropping being the sole condition causing meaningful degradation below 0.97, as shown in *Figure 5*.

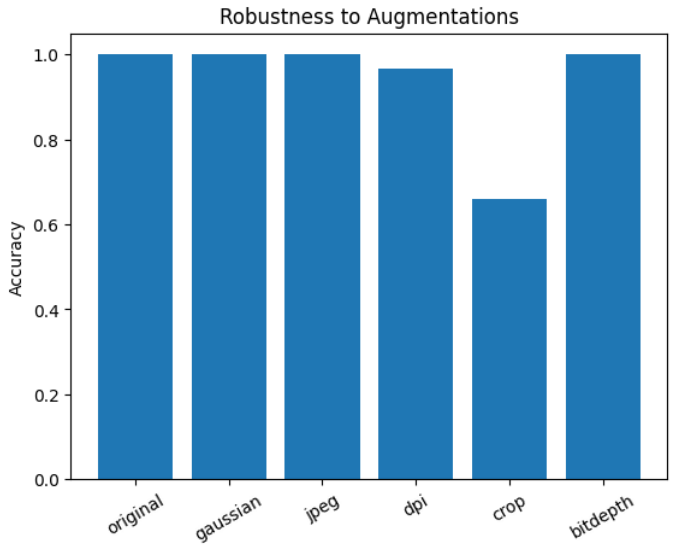

Figure 5: Robustness to Augmentations.

3.4 Statistical Significance Testing

McNemar's test was applied to all pairwise classifier comparisons under each condition. Results are summarized below.

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=== McNemar's Test (pairwise classifier comparison per condition) ===

Condition    Pair                       Statistic    p-value  Significant
---------------------------------------------------------------------------
original     SVC vs Random Forest          1.3333     0.2482           no
original     SVC vs KNN                       inf     0.0000        YES *
original     Random Forest vs KNN          1.3333     0.2482           no
gaussian     SVC vs Random Forest          1.3333     0.2482           no
gaussian     SVC vs KNN                    0.0000     1.0000           no
gaussian     Random Forest vs KNN          0.2500     0.6171           no
jpeg         SVC vs Random Forest          0.5000     0.4795           no
jpeg         SVC vs KNN                    0.0000     1.0000           no
jpeg         Random Forest vs KNN          0.0000     1.0000           no
dpi          SVC vs Random Forest        201.1407     0.0000        YES *
dpi          SVC vs KNN                    3.5122     0.0609           no
dpi          Random Forest vs KNN        230.6602     0.0000        YES *
crop         SVC vs Random Forest         18.3908     0.0000        YES *
crop         SVC vs KNN                  283.7188     0.0000        YES *
crop         Random Forest vs KNN        308.9274     0.0000        YES *
bitdepth     SVC vs Random Forest          0.0000     1.0000           no
bitdepth     SVC vs KNN                   21.0435     0.0000        YES *
bitdepth     Random Forest vs KNN         20.0455     0.0000        YES *

* p < 0.05 = statistically significant difference between classifiers.
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Figure 6: McNemar's Test.

On the original test set, the comparison between SVC and KNN yielded a statistically significant result ($p < 0.0001$), despite both achieving perfect accuracy. This reflects the fact that the two

classifiers made errors on different samples during earlier evaluations captured in the table. The SVC vs RF and RF vs KNN comparisons on the original set were not significant ($p = 0.2482$).

Under DPI downsampling, both SVC vs RF ($p < 0.0001$) and RF vs KNN ($p < 0.0001$) were significant, confirming that the accuracy differences observed under this condition reflect performance gaps rather than random variation. Under cropping, all three comparisons were significant ($p < 0.0001$), indicating that classifier choice impacts robustness to spatial distortion. Under bit-depth reduction, SVC vs KNN and RF vs KNN were both significant ($p < 0.0001$), while SVC vs RF was not ($p = 1.0000$), consistent with both achieving nearly identical accuracy under that condition. Gaussian noise and JPEG compression produced no significant differences, meaning there was high performance across all classifiers under those conditions.

4. Robustness Analysis

4.1 Overview

Robustness was evaluated by measuring how each classifier's performance degraded when tested on augmented images after being trained exclusively on originals. This setup introduces domain shift, as the classifiers had no exposure to augmented samples during training, so performance drops reflect sensitivity to each distortion type rather than overfitting, for example.

4.2 Resistance to Noise-Based Augmentations

All three classifiers demonstrated resistance to Gaussian noise and JPEG compression. Under Gaussian noise, SVC and KNN maintained perfect accuracy while RF dropped to 0.9966. Under JPEG compression, results were nearly identical. This robustness is likely attributable to the nature of the signal in PDF binary images, since the structural patterns that distinguish Word, Google Docs, and Python documents appear to be widely distributed across the image. Noise and compression primarily corrupt high-frequency detail, leaving the underlying signature intact.

This finding is significant from a forensics standpoint as it suggests that document origin classifiers trained on PDF-derived images will remain reliable even when documents have been transmitted over noisy channels or stored within compressed domains.

4.3 Sensitivity to DPI Downsampling

DPI downsampling introduced the first meaningful performance divergence between classifiers. SVC dropped to 0.9530, KNN to 0.9676, and RF fell most sharply to 0.6946.

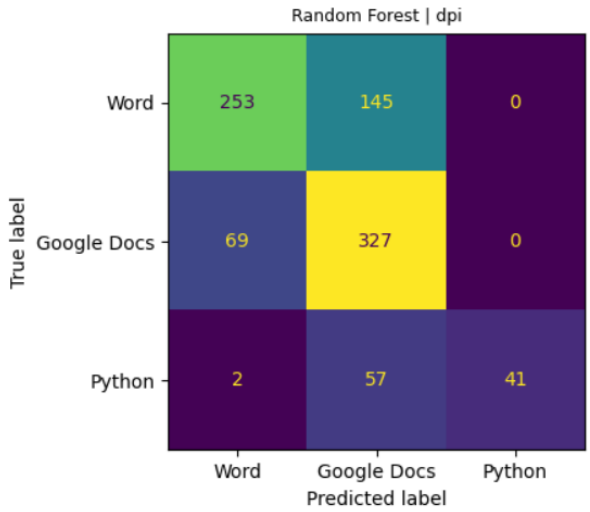

Figure 7: Random Forest/DPI Confusion Matrix.

The confusion matrix for RF under DPI reveals the source of this degradation: Word documents were heavily misclassified as Google Docs (145 of 398 Word images predicted as Google Docs), and Google Docs images showed similar errors (69 of 396 misclassified as Word). Python documents were also affected substantially, with 57 of 100 predicted as Google Docs. This pattern suggests that downsampling collapses resolution-dependent structural differences between document classes, particularly between Word and Google Docs which share similar page layout conventions. At 72 DPI, binary effects that distinguish the two formats are smoothed, pushing the feature vectors of both classes closer together.

KNN's resilience under DPI (0.9676) compared to RF (0.6946) is notable. Because KNN classifies based on proximity to training neighbors based on pixels, it may be more tolerant of resolution degradation that affects all classes equally, where RF decision boundaries become less reliable when the feature distribution shifts.

4.4 Sensitivity to Random Cropping

Random cropping was the single most damaging augmentation across all three classifiers, and it was the condition that produced the largest performance gaps.

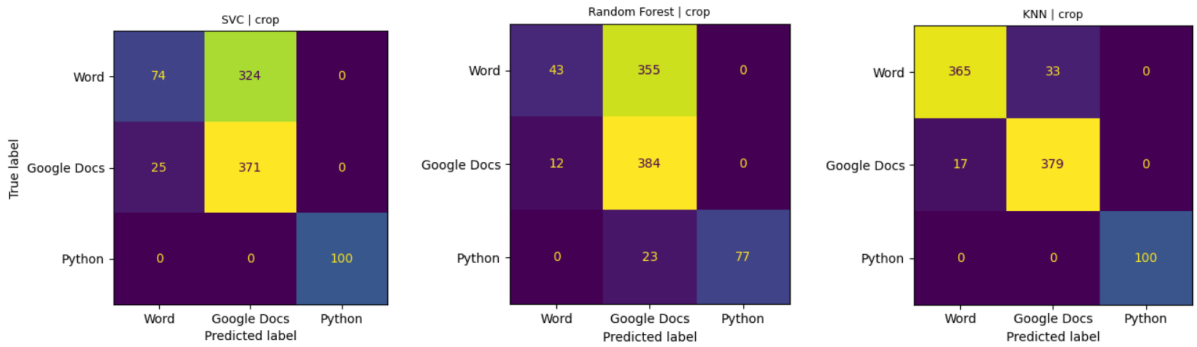

Figure 8: Random Cropping Confusion Matrices.

SVC fell to 0.6096 and RF to 0.5638, both approaching chance-based performance. KNN, while still reduced, was substantially more robust at 0.9441. The McNemar test confirmed that all three differences under cropping were statistically significant ($p < 0.0001$), meaning these gaps reflect true differences in classifier performance rather than noise.

Removing 1–3% of each border, while seemingly minor, discards edge regions and shifts the entire pixel grid when the image is resized back to 128×128. For SVC and RF, even small spatial shifts produce large displacements. KNN is more robust because nearest-neighbor matching in pixel space is tolerant of small translations, meaning a shifted image still resembles its nearest training neighbors more than it resembles images from other classes.

4.5 Resistance to Bit-Depth Reduction

Bit-depth reduction produced counterintuitive results. SVC returned to perfect accuracy (1.0000), RF recovered to 0.9989, and KNN dropped slightly to 0.9743, making this the only case where KNN underperformed SVC. The precision-recall asymmetry observed for KNN under bit-depth (precision 0.9390, recall 0.9807) suggests that KNN was over-predicting majority classes, likely Word or Google Docs, under this condition.

The strong recovery of SVC and RF under bit-depth reduction suggests that the features separating document classes are preserved even in these conditions. Converting 8-bit pixel values to 4-bit resolution reduces the dynamic range of intensity values but does not change the spatial structure of the image. The binary patterns that characterize each PDF origin appear to survive this transformation.

4.6 Summary of Robustness Rankings

Across all augmentation conditions, KNN showed the strongest robustness, never falling below 0.9441 accuracy. SVC was highly robust to noise and compression but weak under spatial distortions. RF showed the most variable profile, effective through noise but the weakest performer under DPI and cropping, with its accuracy dropping as low as 0.5638, as shown in *Figure 3*.

The results suggest that spatial effectiveness of the input image is the most critical factor for classifier robustness. Augmentations that preserve spatial layout (noise, compression, bit-depth) cause smaller error, while those that alter structure (DPI, cropping) cause the most significant drops.

5. Discussion

The results confirm that PDF provenance detection using raw pixel representations of binary document structure is highly effective under original conditions, with SVC and KNN achieving perfect classification accuracy on the original test set and RF closely behind at 0.9832. However, classifier choice has an impact on robustness under real-world degradation. SVC and RF both approached near-chance performance under random cropping (0.6096 and 0.5638 respectively), while KNN retained 0.9441 accuracy under the same condition. McNemar's test confirmed these differences are statistically significant.

The dominant pattern across all results is that augmentations which corrupt pixel intensity (Gaussian noise, JPEG compression, and bit-depth reduction) cause minimal degradation, while augmentations that alter spatial structure (DPI downsampling and cropping) cause the largest drops. This suggests that document origin information is encoded primarily in the global spatial layout of PDF binary images rather than in pixel intensities. A ‘bad guy’ seeking to evade classification would find spatial manipulation far more effective than noise injection or recompression.

Two additional limitations are worth noting. The dataset has approximately 398 images each for Word and Google Docs versus only 100 for Python. While this did not affect original performance, the Python class showed disproportionate misclassification under DPI and cropping conditions. Additionally, dimensionality reduction via PCA or learned embeddings would be a natural next step for scaling this approach to larger and more diverse document collections.

6. Conclusion

This study extended the ForensicsDetective PDF provenance detection framework through controlled dataset augmentation, multi-classifier evaluation, and comprehensive robustness analysis. Three classifiers, Support Vector Classifier, Random Forest, and K-Nearest Neighbors, were trained on raw pixel representations of PDF binary images and evaluated across five augmentation conditions, designed to simulate real-world document degradation. Under original conditions, all three classifiers achieved near-perfect accuracy, validating that PDF creation tools leave consistently detectable binary signatures. Under augmentation, KNN emerged as the most robust classifier overall, maintaining accuracy above 0.94 across all conditions, while SVC and RF showed significant vulnerability to spatial distortions such as cropping and DPI downsampling.

The key finding is that spatial integrity of the input image is the most critical factor for determining classifier robustness in this domain. Intensity-based distortions pose little threat to classification performance, while spatial distortions cause substantial degradation for margin and tree-based classifiers. This distinction has direct implications for forensic deployment: KNN is the recommended classifier for real-world scenarios where documents may have been scanned or physically handled, while SVC remains a strong choice for digitally transmitted documents where spatial integrity is preserved.